

	Assessment:	Project
	Subject:	Business Intelligence 381
	Total:	100

Project Outline:

Socioeconomic Development & Household Well-being in South Africa (SDHSA)

1. Project Overview

The project investigates socioeconomic development, household well-being and economic participation in South Africa. Student teams will work as data analytics consultants supporting a national research and policy institute that requires evidence-based insight for decision-making.

Project challenge

Use a portfolio of related 2024 South African survey datasets to define a defensible analytics problem, select relevant data, engineer an analytical dataset, uncover patterns, build and evaluate suitable models, and communicate a decision-support solution. The project deliberately does not prescribe one mandatory dataset combination or one correct analytical pathway.

The original datasets were downloaded from: <https://isibaloweb.statssa.gov.za/> (Statistics South Africa, 2026). Note that the original data was split into 11 smaller datasets.

2. Project Purpose

Investigate relationships among education, household living conditions, access to basic services, household resources, food well-being, digital connectivity, mobility and economic participation, and translate the findings into an ethical, reproducible and useful business intelligence solution.

3. Required Methodology: CRISP-DM

All project activities must strictly follow the Cross-Industry Standard Process for Data Mining (CRISP-DM). The six CRISP-DM phases remain the methodological foundation, but for assessment they are integrated into four project milestones.

Milestone	Integrated CRISP-DM phases	Primary emphasis
M1 - Business and Data Understanding	(1) Business Understanding (2) Data Understanding	Problem framing, dataset selection, EDA and data-quality investigation
M2 - Data Preprocessing	(3) Data Preparation	Selection, cleaning, integration, feature engineering and analytical dataset construction
M3 - Modelling + Evaluation + Deployment	(4) Modelling (5) Evaluation (6) Deployment	Model design, comparison, business evaluation and deployable analytical product

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M4 -Final Report and Presentation	Final synthesis across all phases	Final report, dashboard, ethics, recommendations, presentation and project review
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4. Data Environment

Teams receive multiple datasets derived from Statistics South Africa 2024 survey data. The datasets vary in analytical grain and purpose. Some are person-level, some household-level, and the labour-market data originate from a separate survey environment. Students are expected to determine which datasets are relevant to their problem and how they may be integrated appropriately.

Data readiness: The supplied datasets must not be assumed to be analysis ready. Teams are responsible for systematically assessing their quality, structure, consistency, completeness and suitability for the chosen analytical problem. Any identified concerns must be supported by evidence, and preprocessing decisions must be justified, implemented reproducibly and validated.

Dataset	Indicative analytical domain
D01_Population_Profile.csv	Population and demographic profile
D02_Learning_Profile.csv	Education and learning profile
D03_Economic_Participation.csv	Economic participation
D04_Dwelling_Conditions.csv	Dwelling and housing conditions
D05_Basic_Services.csv	Water, sanitation and basic services
D06_Household_Energy.csv	Household energy
D07_Household_Wellbeing.csv	Food security and household well-being
D08_Connectivity_Access.csv	Connectivity and digital access
D09_Household_Resources.csv	Income, grants, assets and resources
D10_Mobility_Access.csv	Mobility and access
D11_Labour_Market.csv	Labour-market modelling data

Important data integration requirement

The presence of similar fields across files does not automatically make a row-level join valid. Teams must identify the unit of observation, candidate keys, one-to-one or one-to-many relationships, and cases where aggregation or separate analysis is required. Dataset selection and integration decisions must be justified.

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5. Student Team Responsibilities

- Define a specific decision problem within the broader SDHSA scenario and identify relevant stakeholders.
- Inspect the available datasets and select a defensible subset that addresses the chosen problem.
- Explain why selected datasets are relevant and why excluded datasets are not required.
- Demonstrate the grain, keys and relationships of selected datasets before integration.
- Perform data-quality assessment, cleaning, transformation and feature engineering in R.
- Construct one or more fit-for-purpose analytical datasets without invalid joints or leakage.
- Use exploratory and statistical analysis to establish meaningful patterns and hypotheses.
- Build, compare and evaluate suitable predictive and/or segmentation models.
- Develop decision-support visualisations using Power BI and deploy an appropriate analytical/model component using R/Shiny or an approved R-based approach.
- Integrate ethics, privacy, bias, fairness, transparency and responsible AI considerations throughout the project.
- Maintain reproducible, well-documented code and evidence of team decision-making.

6. Analytical Problems

Teams may pursue different analytical pathways provided the selected problem is supported by the data and the modelling approach is technically appropriate. These are just a few examples. Students are encouraged to explore other analytical challenges and solutions.

Example pathway	Illustrative focus
Household vulnerability	Identify combinations of living conditions, resources, service access and food well-being associated with vulnerability.
Socioeconomic segmentation	Discover meaningful household or population profiles using clustering and interpret the resulting segments.
Economic participation	Investigate factors associated with labour-market participation or employment outcomes.
Digital inclusion	Examine how connectivity, education, geography and household resources relate to digital access.
Service-access inequality	Analyse patterns in water, sanitation, energy, housing or mobility across population groups and geography.

7. Tools and Technical Requirements

- R is the required analytical programming environment. R Markdown or Quarto is strongly recommended for reproducibility.

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- Power BI is required for interactive business intelligence dashboards and visual communication.
- R/Shiny, or another lecturer-approved R-based interface, may be used for deployment of the final analytical/model component.
- Code must be organised, commented, reproducible and run without unresolved errors.
- External research may be used to establish business context, but all external sources must be referenced appropriately.

8. Ethics and Responsible AI Usage

Ethics is not a once-off section added at the end. Teams must identify, document and address ethical considerations throughout the lifecycle, including privacy, inappropriate inference, representativeness, survey weighting, algorithmic bias, fairness, transparency, limitations and responsible communication of results. Consider these ethical issues within the South African context, given that the dataset reflects the country's socioeconomic development, household well-being, and economic participation.

9. Four-Milestone Roadmap

Milestone	Core Deliverables
Milestone 1	Project charter/problem statement; Problem definition document; List of project questions and KPIs; Stakeholder objectives summary; data inventory and selection rationale; EDA; data-quality assessment; initial Power BI exploration; Ethical considerations R and Quarto Markdown scripts for EDA; Summary tables of key variables; Visuals (bar charts, histograms, boxplots); Data dictionary (Excel/PDF)
Milestone 2	Clean and integrated analytical dataset(s); reproducible preparation pipeline; feature engineering; data dictionary; train/test or resampling design; Ethical considerations Cleaned and integrated datasets in R Codebook of transformations <i>Rds</i> , <i>Rmd</i> , <i>Qmd</i> and CSV files of cleaned data
Milestone 3	Candidate and final models; model comparison; technical and business evaluation; deployment strategy; functional R/Shiny analytical/model component. Performance metrics; Evaluation report; Insights summaries; Recommendations; Ethical considerations Interactive Power BI dashboard and Shiny Apps (or other approved R-based interface)
Milestone 4	Final Power BI dashboard(s); comprehensive final report; ethics and responsible AI log; recommendations; executive presentation; final repository and project review; Ethical considerations Final project report (PDF) Group presentation (PPT)

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10. Overall Project Outcomes

- Evidence-based insights on socioeconomic well-being and economic participation in South Africa.
- A defensible integrated analytical dataset created from selected source datasets.
- A validated predictive and/or segmentation model appropriate to the chosen problem.
- An interactive Power BI decision-support dashboard.
- A deployable R-based analytical/model component.
- A professional, reproducible and ethically responsible analytics solution.

11. Academic Integrity and Originality

There is intentionally no single correct dataset combination, model or final answer. Assessment rewards evidence, reasoning, technical correctness, originality and the quality of justification. Reusing a previous project by merely replacing variables or labels does not demonstrate the required learning. All submitted work must represent the team's own analysis and decision-making.

12. Additional Information

- This is a group project, but the group may not exceed three people. Groups will be allocated randomly.
- All work must be original. Copying another group's work will not be tolerated.
- Includes names of all group members on the Cover Page.
- Submit your project milestones electronically on Moodle (BC Connect) before the due dates.
- All writing must be correctly cited and referenced.
- **Plagiarism is a serious offence.** Belgium Campus makes use of software that can scan for plagiarism and penalties apply for any plagiarised work including getting a **0** for the assignment.
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late assignments will not be accepted; missing the deadline is an automatic 0.